POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2016
Programme: BE | Full Marks: 100
Course: Chemistry | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Write the mechanism of buffer action of a solution containing a mixture of methanoic acid and sodium methanoate. 4+4

A buffer solution containing 0.4 mol l^{-1} of ammonia solution and 0.6 mol L^{-1} of ammonium chloride has prepared. What will be the pH of this solution after 0.075 M HCl has been added to the buffer. (assume that volume is unchanged K_b for NH_3 solution= 1.8×10^{-5}

$NH_3+H_2O \Rightarrow NH_4^+OH^-$

b) What do you mean by standard hydrogen electrode? Calculate the emf of the following cell. 4+3

$Zn(s)/Zn^{+2}(0.1M)//Cu^{+2}(1.75M)/Cu(s)$ at 25°C

$E°_{Zn^{++}/Zn}= -0.76V$

$E°_{Cu^{++}/Cu}= +0.34V$

$R=8.314\ Jmol^{-1}K^{-1}$

$F=96500\ C$

2. a) Define Ionization Energy. Why second Ionization Energy is greater than first Ionization Energy? Explain it. Discuss the general trend of its value in the period and group in the periodic table. 1+2+4

b) Give reason 4×2

i. Transition elements form significant number of complexes

ii. s- block elements are reducing in nature

iii. Transition elements show variable valency

iv. Zn^{+2} salts are white

3. a) What are reaction intermediates? Explain the structure and stability of different types of carbocation. 2+2+4

b) Point out the differences between SN_1 and SN_2 reactions with suitable 7

examples.

4. a) What is polymerization reaction? Write the different types of polymerization with suitable examples for each. 2+5

b) Write short notes on: 4×2

i. PVC

ii. Vulcanization of rubber

5. a) Write the procedure involved in mass spectroscopy. How many NMR signals would we expect in compound isomers of C_3H_6O. 5+2

b) Write the functions of lubricants. How cement can be manufactured. 2+6

6. a) "Kathmandu is considered as one of the very polluted cities in Asia". Discuss its causes, effects to human health and ways to control it. 7

b) What are the causes of Hardness of water? How it can be measured in the laboratory. Describe it. 2+6

7. Write short notes on: **(Any two)** 2×5

a) Ozone layer depletion

b) Markovnikoff's rule

c) Thermoplastics and Thermosetting plastics